

END SEM 2026
Foundations of Biomedical Informatics(BIO222)

Max. Marks: 100

Name:

Roll no.:

Time: 90 mins

SECTION-A(15*4 = 60 marks)

1. A rare disease has a highly accurate test, but most positive results are false positives. Why?
A. Test is inaccurate **B. Low prior probability** C. High likelihood D. Sample size is large
2. You want to compare the average heart rate between males and females. Which test is MOST appropriate?
A. Chi-square test **B. t-test** C. F-test D. Anova
3. A model trained on Hospital A EHR data performs well locally but degrades significantly when applied to Hospital B data. What is the most likely reason?
A. The model captured hospital-specific patterns rather than generalizable features
B. The model lacks computational optimization for deployment environments
C. The algorithm choice is suboptimal for cross-institutional datasets
D. The dataset size in Hospital A was insufficient for training
4. A predictive model shows that patients using a specific drug have lower recovery rates. What should be done FIRST?
A. Accept causation **B. Check for confounding** C. Retrain model D. Apply clinically
5. A dataset contains extreme outliers that affect model performance. Which method is MOST appropriate for such a case?
A. One-hot encoding **B. Winsorization** C. Label encoding D. Binning
6. Which technique helps in handling skewed data?
A. Standardization **B. Log transformation** C. One-hot encoding D. Feature selection
7. A CDSS recommends antibiotics based on historical prescriptions. Over time, it reinforces irrational prescribing patterns. This is an example of:
A. Overfitting **B. Feedback loop bias** C. Selection bias D. Regularization
8. Which scenario BEST represents Simpson's Paradox?
A. Trend reverses after aggregating data B. Data has missing values
C. High variance in predictions D. Model overfits training data
9. In SNOMED CT, a concept can have multiple parents (polyhierarchy). What is the main advantage of this?
A. Reduces storage **B. Enables richer clinical representation**
C. Simplifies coding D. Eliminates redundancy
10. A CDSS uses SNOMED CT but fails to capture the temporal progression of disease. What is missing? **Temporal modelling/time series representation**
11. Polyhierarchy simplifies data representation but reduces flexibility. (True /**False**)
12. CDSS decisions are always accurate if trained on large datasets. (True /**False**)
13. Free-text clinical notes can be used directly in CDSS without preprocessing. (True /**False**)

Assertion–Reasoning Questions (*A = Both true, Reason explains Assertion | B = Both true, Reason does NOT explain | C = Assertion true, Reason false | D = Assertion false, Reason true*)

14. **Assertion (A):** Removing all correlated variables from a dataset ensures that the model learns causal relationships.
Reason (R): Correlation and causation are fundamentally different concepts. **D**
15. **Assertion (A):** One-hot encoding can lead to very high-dimensional feature spaces in biomedical datasets.
Reason (R): Each unique category is represented as a separate binary variable, even if some categories occur very rarely. **A**

1. **a.** What are Bayesian Networks? How do they model causal relationships? (Write your answer in only 2 points each.) **b.** Compare Maximum Likelihood Estimate (MLE) and Maximum a Posteriori Estimate (MAP) estimation in the context of clinical prediction systems. (Write your answer in only 2 points each.) **c.** A certain disease affects 1% of the population. A test for the disease is 99% accurate (i.e., the probability that the test is positive given the person has the disease is 0.99, and the probability that the test is negative given the person does not have the disease is 0.99). If a person tests positive, what is the probability that they actually have the disease? **MADHAVI [3+3+4]**

Answer:

a. 1. Definition:

Bayesian Networks are probabilistic graphical models that represent a set of variables and their conditional dependencies using a directed acyclic graph (DAG). Each node represents a variable, and each edge represents a probabilistic dependency between variables. (1.5)

2. Modeling Causal Relationships:

Bayesian Networks model causal relationships through directed edges, where an arrow from one node to another indicates that the first variable has a causal influence on the second. These relationships are quantified using conditional probability distributions, allowing the model to capture how changes in one variable affect others (e.g., Disease → Symptoms → Test Result). (1.5)

Component	Criteria	Marks
Definition	Correctly defines as probabilistic graphical model with nodes & directed edges	1.5
	Partial definition (graph/probability unclear)	0.5 – 1
Causal Relationships	Clearly explains edges represent causal influence	1.5
	Mentions relation but not causal	0.5 – 1
Total		3

b. Maximum Likelihood Estimate (MLE) (1.5)

1. Concept:

MLE estimates model parameters by maximizing the likelihood function, thereby finding parameter values that make the observed clinical data most probable. It relies solely on the available dataset without incorporating any prior assumptions.

2. Application in Clinical Systems:

In clinical prediction systems, MLE is effective when large amounts of patient data are available (e.g., ICU datasets), as it can learn patterns directly from data. However, it may perform poorly when data is sparse or noisy because it does not incorporate prior medical knowledge.

Maximum a Posteriori (MAP) (1.5)

1. Concept:

MAP estimation extends MLE by incorporating prior probability distributions along with the likelihood. It estimates parameters by maximizing the posterior probability, combining both observed data and prior knowledge using Bayes' theorem.

2. Application in Clinical Systems:

MAP is particularly useful in healthcare settings where prior medical knowledge (e.g., known disease prevalence, expert guidelines) is available or when data is limited. It helps produce more robust and stable predictions by preventing overfitting to small datasets.

c.

Step	Criteria	Marks
Identify Probabilities	Correct values (0.01, 0.99, etc.)	1
Formula	Correct Bayes' theorem expression	1
Substitution & Calculation	Correct working	1
Final Answer	Correct result (0.5 or 50%)	1
Total		4

2. A machine learning-based CDSS predicts the risk of sepsis in ICU patients. While the model has high accuracy, clinicians are reluctant to use it because it behaves like a “black box.” **a.** Why is explainability critical in healthcare CDSS? **b.** Explain how feature importance or SHAP values can improve interpretability. **c.** What are the risks of deploying a non-explainable model in critical care? **d.** Suggest two design strategies to balance accuracy and interpretability

VARSHA [2+3+3+2]

a) Why is explainability critical in healthcare CDSS?

- Clinicians need to trust and understand the model's recommendations before using them in patient care.
- It ensures accountability and transparency, especially in high-risk decisions like ICU management.

b) How do feature importance or SHAP values improve interpretability?

- Feature importance identifies which variables (e.g., vitals, lab values) most influence predictions.
- SHAP values quantify the contribution of each feature to an individual prediction.
- They help clinicians understand why a prediction was made, improving trust and validation.

c) Risks of deploying a non-explainable model in critical care

- Lack of trust: Clinicians may ignore or misuse the model.
- Hidden bias/errors: Model may rely on incorrect or biased features without detection.
- Patient safety risk: Wrong predictions without explanation can lead to harmful decisions.

d) Two design strategies to balance accuracy and interpretability

- Use interpretable models (e.g., decision trees, logistic regression) where possible.
- Apply post-hoc explanation methods (e.g., SHAP, LIME) to complex models.

3. A hospital implements an AI model to predict disease risk using patient imaging and clinical data. After deployment, several issues arise: the model performs poorly with patients of darker skin tones. It shows high accuracy but fails in real-world clinical settings. Doctors blindly trust the model outputs. **a.** Identify and explain any three types of AI failures illustrated in this scenario. **b.** Why is “correlation ≠ causation” a critical issue in medical AI? **c.** What is meant by the “halo effect” in AI models, and how does it affect decision-making? **d.** Suggest three measures to ensure safe and ethical deployment of AI in healthcare.

NEELIMA

[3+2+2+3]

Answer: a. Identify and explain any three types of AI failures illustrated in this scenario.

Any 3

Bias / Algorithmic bias (skin tone bias)

→ Model underperforms on darker skin tones due to non-representative training data.

Poor generalization / External validity failure

→ High accuracy in training/testing but fails in real-world clinical settings (dataset shift).

Overfitting / Lack of robustness

→ Model memorizes patterns in training data but cannot adapt to new populations.

Automation bias (doctor over-reliance)

→ Clinicians blindly trust model outputs without critical evaluation.

Data distribution shift / Domain shift

→ Differences between training and deployment environments degrade performance

b. Why is “correlation ≠ causation” a critical issue in medical AI?

- Correlation means two variables are associated, but it does not imply that one causes the other. A third hidden factor (confounder) may influence both.
- Machine learning models are designed to optimize prediction, not understand causality. So they may pick up statistical patterns that are not biologically meaningful.
- A model might associate certain scanner types, hospital IDs, or skin tone distributions with disease risk.
- Or it may predict worse outcomes for patients receiving a drug, not because the drug is harmful, but because sicker patients are more likely to receive it (confounding).

c. What is meant by the “halo effect” in AI models, and how does it affect decision-making?

The halo effect is a cognitive bias in which users (e.g., clinicians) place excessive trust in an AI system because it appears accurate, advanced, or has performed well in the past, and therefore assume all its outputs are reliable.

Impact on decision-making:

- Automation bias: clinicians defer to the AI instead of critically evaluating it
- Reduced vigilance: contradictory evidence (symptoms, tests) may be ignored
- Error amplification: wrong predictions go unchallenged and can directly influence care
- Risk in edge cases: failures in underrepresented groups or rare conditions are missed because trust remains high

d. Suggest three measures to ensure safe and ethical deployment of AI in healthcare.

Measures for safe and ethical deployment of AI in healthcare:

1. **Bias mitigation through diverse data:** Use representative, multi-center datasets covering different demographics to ensure fair performance across all patient groups.
2. **Human oversight (human-in-the-loop):** Ensure AI acts as a decision-support tool, with clinicians reviewing and validating outputs to avoid automation bias.
3. **Continuous validation and monitoring:** Perform external validation before deployment and ongoing performance audits in real-world settings to detect drift, errors, or bias.

4. A healthcare network integrates patient data from multiple hospitals. Each hospital uses different coding systems, leading to inconsistencies in the representation of diagnoses and treatments. An ontology-based framework is proposed to unify the data. **a.** How does ontology enable semantic interoperability across healthcare systems? **b.** What are two challenges in mapping different terminologies into a unified ontology? **c.** Explain the role of standard ontologies (e.g., SNOMED CT) in this process. **d.** How can ontology improve data-driven decision-making and analytics? **MANSI** **[3+2+2+3]**

a) Ontology helps different healthcare systems “understand” each other by providing a common meaning for medical terms. Since hospitals may use different coding systems, the same condition can be recorded in multiple ways. Ontology maps these different terms to a single standard concept, so the meaning remains consistent across systems. It also defines relationships between concepts (for example, a disease and its symptoms or treatments, which helps systems not just exchange data, but also interpret it correctly. This allows smooth data integration and reduces confusion caused by different terminologies.

b)

1. **Different levels of detail:** Some systems record very specific information, while others use more general terms, making mapping difficult.
2. **Ambiguity in terms:** The same word can mean different things in different contexts, or different words can mean the same thing, leading to errors in mapping.

c) Standard ontologies such as SNOMED CT provide a **uniform set of medical terms and codes** that can be used across all healthcare systems. They help in maintaining consistency and make it easier to **share and compare data** between hospitals. They also organize medical concepts in a structured way, which helps in better classification and understanding of clinical information.

d) Ontology improves decision-making by making data more **consistent and reliable**. When all systems use a common structure and meaning, the data becomes easier to analyze. It also helps in combining data from different sources, which gives a more complete picture of the patient. This is useful for clinical decision support systems, predictive models, and identifying patterns in diseases or treatments. Overall, ontology leads to **better insights and more accurate healthcare decisions**.